

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence* (collective phase order), *compatibility* (local descriptions agree where their domains overlap) and *self-representation* (the state carries an accurate reconstruction of itself). Coherence supplies neither of the others, and five results establish that with no cortical hypothesis. Synchrony bounds content disagreement only through encoders shared on overlaps, and then, uniformly over pairs, only inside a patch small enough for the bound to carry information at all. That grain belongs to the uniform bound rather than to phase order: counting the pairs that can disagree leaves a scale-free fraction naming none of them, chaining patches through the sites they share carries the bound a distance linear in the hops travelled, and at a frequency-locked configuration the coupling graph’s algebraic connectivity replaces the population count outright—each against a declared input that coherence itself does not supply. The resultant does not determine the state: a winding field and a uniform one are stationary at one coupling, at opposite ends of that observable. A unique fixed point is not a self-model,

since a contraction can discard its input: accuracy across a family of states requires distinguishable codes. Where the encoding is a region's restriction, that resolution is demanded of the region before any readout is chosen. And agreement about a change takes communication. Four of the five are checkable for a machine candidate, which fixes what a claim of machine unity or self-representation commits its maker to measuring. Under stated hypotheses, compatible local states then glue to a unique global state that an accurate contracting reconstruction fixes uniquely. Installed energy bounds the mean coupling a declared actuator supplies, so a coherent stationary branch requires it to exceed $2D/\kappa$; κ is derived here for neither cortex nor a processor. A Lean 4 development carries these results and their composition, with the eight connecting hypotheses as explicit arguments, no incomplete proofs, and an audited axiom footprint no wider than its libraries'. Further controls bound what observation can decide: distinct mechanisms share stationary phase distributions, and coherent adaptation can lose the structure it was meant to acquire. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, which a within-subject emergence-from-anaesthesia design can test; the finite-size estimate here is biased below 1/2 by its escape criterion, as a deterministic reference confirms. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling. Testing it requires calibrated geometry, resolved phase dynamics and validated decoders.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful

7 coordination would still leave a representational question: what makes the
8 coordinated activity a description of one situation, with a perspective on
9 that situation?

10 We distinguish three requirements. *Coherence* concerns collective phase
11 order. *Compatibility* concerns whether local descriptions agree where their
12 domains overlap. *Self-representation* concerns whether the resulting state
13 carries an accurate internal reconstruction of itself. We propose the latter
14 two relations as formal correlates of conscious unity and a minimal self; their
15 interpretation requires evidence beyond the phase model.

16 Unity has been approached differently elsewhere: integrated information
17 theory treats integration and exclusion as constitutive (Tononi et al., 2016),
18 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene
19 et al., 2014), and predictive-processing accounts derive both from a hierar-
20 chical generative model (Friston, 2010). Here the three requirements are
21 kept apart rather than derived from one another, so that each carries its own
22 assumptions and admits its own measurement; Section 10.3 returns to the
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand’s relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions

30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-
34 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
35 overlaps convert phase order into a bound on content disagreement. Compat-
36 ible spatial measures have a unique global extension; approximate agreement
37 controls a specified global selection. A contracting reconstruction makes that
38 extension its unique fixed point only if it accurately reconstructs the exten-
39 sion. Accuracy across a family of states imposes a further condition on the
40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
42 Stationary phase distributions can be shared by stronger coupling, weaker
43 diffusion and a matched common drive. Finite ramps change the observable
44 onset. A winding field is stationary at the same coupling as a uniform one and
45 has the opposite resultant, so that observable does not determine the state.
46 Coherent adaptation can lose alignment with a designated environmental
47 template, and a compatibility estimator needs diagnostics that distinguish
48 content from decoder artifacts. These findings determine the measurements
49 needed to test the proposed connections.

50 Three of the ingredients are standard: the synchronization threshold
51 $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing
52 of compatible local sections, and the Banach fixed-point theorem (Banach,

1922). Four things here are not. The installed-energy condition makes coupling a purchased quantity: a declared actuator’s stored energy bounds the mean coupling it supplies, so coherence on the stationary branch carries a resource requirement in units a measurement can reach, and for cortex that requirement reduces to a figure the mode prices must deliver. The coherence-to-content bound converts phase order into a statement about descriptions rather than about timing, and its patch-local form is what keeps it from being vacuous in a large population. A stability criterion on the reweighted coupling kernel, together with an existence condition on an amplitude layer, settles when a low resultant reports weak coupling and when it reports a winding field. And the composition is machine-checked together with a witness discharging all eight hypotheses at once, which is what makes their conjunction non-empty rather than a list no object satisfies.

Granting those premises, collected in Table 1, the account runs as follows. Extracellular contraction on awakening raises endogenous field amplitude and with it the effective coupling between cortical oscillators. Once that coupling passes $2D$, a coherent phase branch exists, and from an initial spread within a quarter turn the population settles onto it; other initial data leave windings stationary at the same coupling. Encoders shared where cortical territories overlap turn that phase order into agreement about the quantities the territories describe in common, so the local descriptions glue: there is one state of which each is a restriction. Part of that state encodes the state itself, and a readout of that part reconstructs it accurately enough, and

76 sensitively enough across the states that matter, to make it the unique state
77 the system models itself as being. On this account the glued state is what
78 makes distributed activity one situation, and its accurate self-reconstruction
79 is what gives that situation a perspective. Every step of it is conditional.

80 Section 2 collects the results that hold whatever physical field is proposed
81 and whatever substrate carries it; a reader who rejects the cortical hypothesis
82 of Section 9 still has to accommodate them. Sections 3–5 then develop the
83 three requirements together (Fig. 1), and Section 6 reads the unconditional
84 results against a machine candidate. Section 7 composes them, stating the
85 thermodynamic constraints and all eight connecting hypotheses per branch;
86 its supporting models of feedback, learning and resource supply have distinct
87 state spaces and assumptions. Section 8 then examines what the observables
88 can establish, before Section 9 develops the cortical hypothesis and an awak-
89 ening experiment. The hypothesis is that extracellular contraction increases
90 field-mediated coupling; its direction, rate and relation to recovery require
91 independent calibration. Supplemental Material organizes the constructions
92 and controls behind the claims they support. Table 2, at the end, lists the
93 symbols and the section that defines each.

94 **2 What holds without the cortical hypothesis**

95 Most of what follows is conditional. Table 1 names eight hypotheses that
96 the composition consumes rather than derives, and the cortical identification

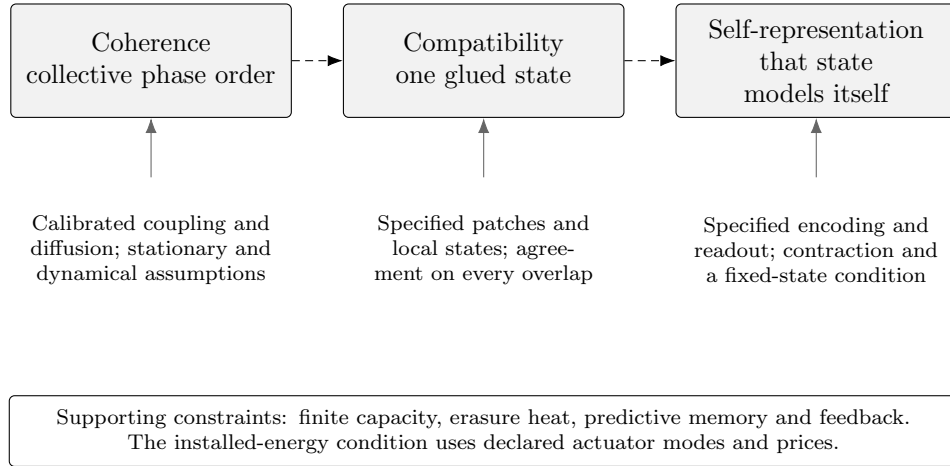


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

97 of Section 9 is a further commitment on top of them. Five results are not
 98 conditional in that way. They constrain any account that reads unity off
 99 phase coherence, whatever physical field is proposed to carry the coupling
 100 and whatever substrate carries the field. Each is stated below where it is
 101 used; collected here, they are the article’s fixed points.

102 **Phase order constrains content only inside a patch too small to**
 103 **carry unity.** Assign every patch the same phase and let two local mass
 104 profiles disagree at a shared site. The profiles are chosen independently of
 105 phase, so synchrony places no constraint on their disagreement, and they

106 have no common extension: perfect locking is consistent with local descrip-
 107 tions that do not glue (Section 4). What the positive result of Section 4.2
 108 costs is exactly this gap. A bound on content disagreement needs encoders
 109 already related on overlaps, and that relation is a supplied fact about the de-
 110 scriptions rather than a consequence of the dynamics. Supplying it relocates
 111 the limit rather than lifting it. Eq. (6) bounds every pair at once and a chord
 112 never exceeds 2, so at the patch-local order of the sheet in Section 8.3 that
 113 uniform reading carries information only at 3 sites or fewer. The grain is the
 114 uniform reading's and not phase order's, and Section 4.2 states three results
 115 without it: a scale-free count of the pairs that can disagree, which names
 116 none of them; a bound chained through the sites a cover's patches share,
 117 which travels a distance linear in the hops taken; and, at a frequency-locked
 118 configuration, a bound read off the coupling graph's algebraic connectivity,
 119 which carries no population count at all. What none of the three changes is
 120 the constraint stated here: a count locates no site, and the other two con-
 121 sume a declared cover or declared hardware data, so phase order by itself
 122 still says nothing about territories too far apart to share a patch, which is
 123 where the compatibility claim is made.

124 **The order parameter does not determine the state.** A field wind-
 125 ing once across a periodic sheet has resultant $r = 0$ — the order parameter
 126 of Eq. (1) below — while every patch of it stays locked, and this is not a
 127 contrived initial condition. On a periodic array of any dimension, a winding

128 is stationary for *any* isotropic symmetric kernel at identical frequencies, be-
 129 cause the coupling term is odd in the separation against a kernel even in it
 130 (`char_is_kuramoto_trajectory`). Section 8.3 exhibits a sheet on which uni-
 131 form and winding initial conditions reach 0.9657 and 0.0083 at one coupling,
 132 diffusion and kernel. A low resultant therefore reports either weak coupling
 133 or a winding field, and no measurement in r alone separates them. The state
 134 also carries an integer the resultant does not: read around a ring, the same
 135 winding has winding number q , and a nonzero q obstructs every global real-
 136 valued phase lift (`winding_degree_obstructs`). The resultant is an average
 137 over sites and moves when they are reweighted or the patches redrawn; the
 138 degree telescopes and does not.

139 **A unique fixed point is not a self-model.** A constant readout con-
 140 tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
 141 is consequently the wrong criterion, and Section 5.2 gives the right shape
 142 of one: reconstructing M states pairwise more than 2ε apart within error ε
 143 requires M distinguishable codes, so an alphabet of 2^b codes caps the family
 144 at $M \leq 2^b$. This is a counting argument and holds for any encoder and
 145 readout whatever.

146 **Where the encoding is a restriction, the region must resolve the**
 147 **family.** Read the code as the state's own restriction to a region and
 148 the criterion moves off the readout entirely. A boundary whose region
 149 reads two relevant states identically cannot reconstruct them more than 2ε

150 apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant
151 boundary that does reconstruct a separated family restricts injectively on
152 it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
153 region sits is fixed before an encoding is chosen, so no readout supplies
154 resolution the region lacks.

155 **Agreement after a change takes communication.** In a synchronous
156 network whose updates read only the site’s own state and its declared neigh-
157 bours’ messages, two configurations agreeing on a site’s causal past through
158 T rounds leave that site identical at round T . Two interventions outside
159 that past that require reports more than 2ε apart therefore cannot both be
160 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
161 ity and timely compatibility are therefore different requirements, and cutting
162 the path removes the second while leaving the first intact.

163 None of these depends on the coupling being electromagnetic, on the sub-
164 strate being cortex, or on the eight hypotheses of Table 1 holding. Section 6
165 applies four of them — all but the one about the resultant, which is a state-
166 ment about a spatial observable — to a candidate that is neither biological
167 nor field-coupled.

168 3 Coherence: physical coupling and its re- 169 source condition

170 3.1 Dynamics and the stationary threshold

171 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

172 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
173 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
174 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
175 quency heterogeneity and diffusion are distinct parameters. The stationary
176 calculation below uses identical frequencies and this scalar coupling. Ap-
177 plying it to a spatial kernel requires an aggregation rule and a check of the
178 resulting approximation.

179 For identical frequencies in their rotating frame, the positive normalized
180 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

181 where I_n are modified Bessel functions and ψ is the collective phase. The for-
182 mal derivation uses the absence of stationary current for a periodic gradient
183 drift to obtain this density, then verifies normalization and self-consistency.

184 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above
 185 threshold, exactly one positive solution coexists with zero and increases
 186 strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

187 Figure 2 shows this stationary branch.

188 Linearization about the uniform density gives rate $K/2 - D$ for the first
 189 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 190 exactly above threshold. About the coherent density, the quadratic form
 191 is strictly dissipative transverse to collective rotation, whose rate is zero.
 192 These linear estimates concern the limiting equation. Relating them to finite
 193 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 194 tion; exchange symmetry alone does not establish the needed independence
 195 (Sznitman, 1991).

196 **3.2 Installed energy bounds mean coupling**

197 A spatial model replaces the finite sum by integration against a kernel. One
 198 declared implementation carries response modes with continuous profiles ϕ_i ,
 199 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_\Xi(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

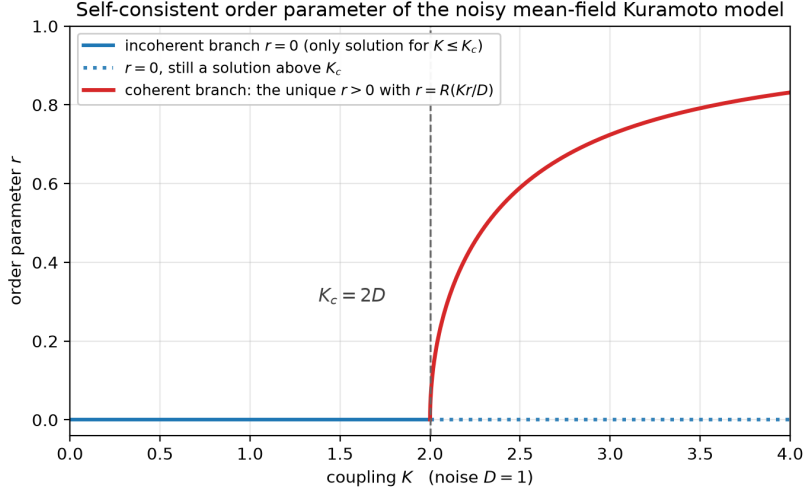


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

200 The same configuration sets the reciprocal kernel and stored energy. An
 201 executed transition changes the kernel through its changed occupancies and
 202 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 203 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 204 pair reconstruction on refining spatial partitions converges to its product-
 205 measure integral under the specified compactness and sampling assumptions;
 206 the refinement index is independent of the number of control updates.

207 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 208 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

209 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned}
 \tag{5}$$

210 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 211 coherent stationary branch and any growing Fourier mode about the uniform
 212 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 213 is necessary: energy above it need not produce sufficient coupling because
 214 the bound can be loose.

215 The quantity constrained is installed energy rather than dissipated heat,
 216 and the condition binds through κ rather than through the energy, since
 217 any macroscopic substrate carries installed energy far above the thermal
 218 floor. Read forwards it therefore excludes no macroscopic candidate. Read
 219 backwards it is a number. Take the region the field arithmetic takes, a
 220 sphere of one LFP decay length holding 1676 neurons, at the metabolic rates
 221 Supplemental Material collects and over one phase-diffusion time $1/D = 2$ s
 222 at $D = 0.5$ rad²/s: the installed energy is 1.0×10^{-6} J, and the condition is
 223 met at every κ above 9.6×10^5 J⁻¹s⁻¹. That figure is what an identification
 224 of cortical mode prices has to deliver, and the first identification delivering
 225 less refutes the coupling-gated account for cortex at this region and interval.
 226 Another interval rescales the figure by its ratio to this one.

227 What the condition discriminates, it discriminates through that factor.

228 The required κ scales as D^2 and inversely with the region's cell count, so at
 229 fixed mode prices the condition ranks candidate substrates by their installed
 230 energy; in a declared implementation it decides outright, the learning witness
 231 of Section 7 clearing the line after one update while its matched control stays
 232 on the exclusion side at the same conversion factor and diffusion. What is
 233 missing for cortex is a price list rather than a mechanism. A mode is a priced
 234 rank-one contribution to the kernel, cortical field generation has no canonical
 235 decomposition into such modes, and choosing the basis would choose the
 236 answer. Applying the scalar threshold to heterogeneous or signed spatial
 237 dynamics requires a justified reduction, and identifying the installed kernel
 238 with a cortical field is a separate physical commitment.

239 **4 Compatibility: from coordinated activity** 240 **to a global description**

241 **4.1 Why common timing is insufficient**

242 For overlapping patches U and V , compatibility means that restricting either
 243 local description to $U \cap V$ gives the same object. Descriptions can agree on
 244 shared quantities while contributing different information elsewhere. Com-
 245 patible sections of the specified sheaf have a unique global extension. On
 246 a finite cover, local finite spatial measures agreeing as measures on every
 247 overlap also have a unique finite-measure extension whose sheaf image is the

248 glued section. No mass-one normalization is imposed.

249 In the cup example, visual and reaching descriptions must express a
250 shared quantity, such as cup location, in common coordinates. Restriction
251 retains that quantity while dropping details particular to each description.
252 Agreement permits surface appearance and hand configuration to contribute
253 to one situation. The mathematical counterexample assigns the same phase
254 to every patch while two local mass profiles disagree on a shared site. Those
255 profiles have no common extension. Because they are chosen independently
256 of phase, synchrony places no constraint on their disagreement.

257 This distinction limits the biological interpretation. Receptive territories
258 and content domains must be related to spatial patches, and the descriptions
259 must support the required extension. Matching overlap marginals of de-
260 coded probability laws need not determine a unique global joint distribution
261 (Abramsky & Brandenburger, 2011).

262 The cover is itself a choice, and the construction constrains it in one way
263 only: synchronized patches agree where they overlap. Where they do not
264 overlap there is nothing to compare, so a cover by pairwise disjoint patches
265 satisfies that obligation for *every* assignment of local descriptions and phases
266 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-
267 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-
268 ship of the class then reports nothing about the system beyond the patches
269 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy
270 the theorem does not assume: that distinct patches meet (`HasNonempty-`

271 Overlaps), and that what they share is a quantity fixed independently of
 272 the agreement it is being used to establish.

273 4.2 When coherence does constrain content

274 A declared relation between phase and content gives a positive result. Let
 275 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 276 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 277 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 278 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 279 sites; their equality on overlaps is a supplied relation between descriptions of
 280 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 281 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 282 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

283 Here discrepancy is the norm of the difference, and r and N are read on
 284 whichever population the bound is applied to. That choice is what decides
 285 whether the estimate says anything. A patch carries an order parameter of
 286 its own, and reading the bound there carries that patch's resultant and that
 287 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
 288 `coherence`), so a large population observed through small patches is con-
 289 strained by the small count. Read instead on the whole array, the factor N

290 makes the estimate weak, because it draws a pointwise guarantee from an
 291 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
 292 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
 293 tion permits disagreement even at perfect locking. A common nonexpansive
 294 linear projection preserves agreement, but agreement of a projection does
 295 not establish agreement of the full vector.

296 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
 297 constrains anything only where $\sqrt{2} N \sqrt{1 - r^2}$ falls below that. The winding
 298 sweep of Section 8.3 fixes both ends of that range on one sheet. Read on the
 299 sweep's own patch, 625 sites at patch-local order 0.9015, the bound is $382.5 L$
 300 and constrains nothing; holding that order fixed, the count alone brings it
 301 below the maximum chord only at 3 sites or fewer. A nearest-neighbour pair
 302 of the same winding, the smallest patch there is, gives $0.1388 L$. Shrinking
 303 patches therefore thins the overlaps at the same rate as it tightens the bound,
 304 and the two ends are not reached by the same cover. At a columnar grain
 305 the estimate constrains agreement inside one locally synchronous population
 306 and is vacuous across the distant territories whose agreement unity is about.

307 That vacuity is the uniform bound's, and three results decline the step
 308 that puts N there: a sum of N^2 nonnegative terms bounded by its total,
 309 which permits all the disorder in a population to sit in the pair the conclu-
 310 sion is about. Asking instead how *many* pairs can be far apart is Markov's
 311 inequality on the same sum, and its answer carries no population count: at
 312 order r the fraction of pairs separated by a chord of c or more is at most

313 $2(1 - r^2)/c^2$ at any size of population (`chord_fraction_le`), and the same
 314 sum read by rows bounds the fraction of *sites* that are far from a fixed frac-
 315 tion of the population (`site_fraction_le`). Read on the patch above, where
 316 the uniform bound gives $382.5 L$ and constrains nothing, this gives a fraction
 317 at most 0.37 of pairs disagreeing by more than L , and it improves like the
 318 square of what the uniform bound improves like. It is a different quantifier
 319 and not a stronger statement: a count names no pair, the exceptional set
 320 it permits is unlocated and may be exactly the sites a cover's overlaps sit
 321 on, and a guarantee about a named overlap therefore still needs the uniform
 322 bound at its own grain.

323 Distance is reached by chaining rather than by counting. Index the cover
 324 by its nerve, in which two patches are adjacent when they share a site. The
 325 chord is a metric, so a bound β on the separation inside every patch composes
 326 along a walk: a walk of k hops bounds its endpoints by $(k + 1)\beta$, the count
 327 being $k + 1$ and not k because both endpoint sites pay a step inside their own
 328 patch while the shared sites are interior to the chain (`chord_le_of_patch_`
 329 `walk_coherence`), and the content statement follows with it for any cover
 330 whose nerve has diameter k (`compatible_of_patch_nerve`). No patch need
 331 contain both sites, which is the situation of every pair of territories far enough
 332 apart, so shrinking patches and spanning territories are two ends of one chain
 333 rather than two ends of one bound. What it costs is that the accumulation is
 334 linear and nothing makes it sublinear: at the sheet's nearest-neighbour value
 335 $\beta = 0.1388 L$ the chain stays inside the maximum chord for 13 hops, which

336 is 14 patch diameters and 0.219 mm of that sheet. Connectivity of the nerve
 337 is a property of the cover, supplied rather than derived, and a cover whose
 338 nerve is disconnected supports no conclusion between its components.

339 The third result reads no resultant at all. Where a configuration is
 340 frequency-locked to a common rotation Ω , its phase differences are not free:
 341 each site's detuning is balanced by its coupling term, $\omega_i - \Omega = \sum_j K_{ij} \sin(\theta_i -$
 342 $\theta_j)$, which is the Kuramoto equations at a rigidly rotating solution rather
 343 than a further postulate. Pairing that balance with the configuration itself,
 344 symmetrizing, and applying Jordan's inequality termwise turns it into the
 345 coupling graph's Dirichlet form, whose algebraic connectivity λ_2 then bounds
 346 the spread about the mean phase $\bar{\theta}$ by $2\lambda_2 \|\theta - \bar{\theta}\| \leq \pi \|\omega - \Omega\|$ (`spread_le_`
 347 `of_frequency_locked`). Read at a named pair it gives a content residual
 348 of $L\pi \|\omega - \Omega\| / (\lambda_2 \sqrt{2})$ with no population count, no patch and no cover ge-
 349 ometry in it (`chord_le_of_frequency_locked`, `compatible_of_frequency_`
 350 `locked`), and it improves as the coupling graph becomes better connected,
 351 which is the dependence a resultant cannot express. Four things are declared
 352 and none is discharged: λ_2 is hardware data in the standing of κ ; the locked
 353 configuration is assumed, and nothing here proves one exists; the quarter-
 354 turn confinement is assumed and is not implied by locking; and the detuning
 355 enters in the population's ℓ^2 norm, so a population whose frequency spread
 356 grows with its size pays for that growth through the data. What is removed
 357 is the unconditional factor, not the physics of heterogeneity.

358 Structural connectivity is a declared graph and is what supplies λ_2 . Mea-

359 sured functional connectivity supplies it no better: it is an estimate, so it
 360 inherits the decoder failure modes of Section 8, a shared prior shrinking es-
 361 timates towards agreement and regional bias manufacturing disagreement,
 362 and feeding an estimate to a hypothesis strengthens no conclusion drawn
 363 from it. Using it to choose the cover is worse than neutral, being the move
 364 of Section 4 that empties a gluing claim, in different clothes.

365 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 366 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 367 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 368 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

369 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 370 duce an initial mismatch. Approximate observations leave a nondecaying
 371 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 372 update to the phase process.

373 4.3 What approximate agreement determines

374 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 375 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 376 Weights that vanish outside their patches and sum to one at each site select
 377 a global profile by weighted averaging. The selection lies within ε of each
 378 patch there, and any two such selections differ by at most ε , independently

379 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 380 $L\varepsilon$ across selections.

381 At zero tolerance, the selection is the unique exact extension. At every
 382 positive tolerance there are families with no exact extension; selection then
 383 gives an approximation whose dependence on weights is bounded. Supple-
 384 mental Material gives a three-site witness attaining the bound. The mass-
 385 profile construction uses the scalar case of the content dynamics and addi-
 386 tionally requires nonnegativity of the initial profiles and observations. The
 387 general vector result supplies no identification with the spatial-measure sheaf.

388 These constructions concern declared objects: a specified cover, contents
 389 valued in a normed space, and encoders related on overlaps. They select
 390 neither the neural cover nor the decoder, and identifying their contents with
 391 perceptual content, or establishing that the identification bears on experi-
 392 ence, is a further modelling task; Section 8 sets out what a measurement of
 393 it would have to establish.

394 **5 Self-representation: a state that models it-** 395 **self**

396 **5.1 The reflexive construction**

397 Compatibility gives a global state. Reflexivity relates it to an encoding within
 398 that state. Equip global sections with a complete metric. If the additional

399 restriction-resonance condition identifies encoding with restriction to a region
400 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

401 The region and readout are fixed across the state space. If F contracts and
402 fixes the section s_* formed from the local descriptions, Banach's theorem
403 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
404 plies the contraction and $F(s_*) = s_*$; restriction resonance is separate. Su-
405 percritical coupling makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$
406 less than one for a relaxation time $\tau > 0$. Constructing a map with that
407 factor remains a modelling obligation.

408 In the cup example, the global state includes the represented scene and the
409 perceiver's relation to it. A local encoding must retain enough information
410 for the readout to recover that modelled state; cup position alone would be
411 insufficient if hand configuration and viewpoint could not be reconstructed.

412 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
413 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

414 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
415 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
416 weakens.

417 5.2 Reconstruction across a family of states

418 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 419 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 420 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

421 Thus M relevant states pairwise more than 2ε apart require at least M
 422 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 423 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 424 existing self-prediction map.

425 This is an additional accuracy criterion on a family, with no requirement
 426 for multiple exact fixed points. A blind encoder or constant readout fails it
 427 for separated states despite having a unique fixed point.

428 Where the encoding is read as the state's own restriction to a region, the
 429 criterion becomes a condition on the region. Restriction resonance requires
 430 the encoder and that restriction to identify the same pairs of states, so a
 431 boundary reconstructing two relevant states more than 2ε apart while its re-
 432 gion reads them identically is not resonant (`not_isRestrictionResonance_`
 433 `of_reconstructs`); read forwards, a resonant boundary that reconstructs
 434 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
 435 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
 436 cally is fixed by where the region sits, before an encoding is chosen, so the

437 resolution the family demands is the region's and no readout supplies it. The
 438 implication runs one way: a region that does resolve the family is not thereby
 439 shown to carry a resonant encoding, and E89 supplies no such resolution.

440 Those two demands are jointly satisfiable with nothing folded anywhere.
 441 Take the region to be the whole substrate: restriction is then the identity, so
 442 the boundary is resonant, reconstructs every declarable family exactly, and
 443 restricts injectively on all of them (`topAvatar_satisfies_obligations`). A
 444 properness condition on the region (`IsProperAvatar`) is therefore a separate
 445 requirement rather than a consequence of resonance and resolution, and what
 446 a fold into a local region is asked for. Its quantitative form replaces the
 447 alphabet by a metric. With an L -Lipschitz readout, a family reconstructed
 448 within ε whose members lie pairwise more than $d_{\min}$ apart has δ -separated
 449 codes whenever $\delta L \leq d_{\min} - 2\varepsilon$, so the family is no larger than the δ -packing
 450 number of what the region carries (`encard_le_packingNumber_restrict-`
 451 `ToAvatar`). That is the shape a continuous family of states needs, and at
 452 the whole substrate it says nothing, because the region's range is everything.
 453 The gain L is declared and derived here for no physical readout, and at
 454 unbounded gain the bound says nothing.

455 5.3 Agreement after a change takes communication

456 Snapshot compatibility does not specify how quickly descriptions respond
 457 when a shared quantity changes. In a synchronous network whose updates
 458 read only the site's own state and declared neighbours' messages, two initial

459 configurations agreeing on a site’s causal past through T rounds produce the
460 same state there at round T . The deadline consequence is the one Section 2
461 states, and it follows from the reconstruction bound in Section 5.2: a report
462 from that site is a readout of a code that the two interventions leave identical.

463 A three-site delay line witnesses failure after one round and exact report-
464 ing after two; cutting the path prevents delivery at every round. A report
465 may still happen to be correct about one fixed world. The bound concerns
466 responsiveness across interventions.

467 Throughout this section *prediction* names a reconstruction map, and the
468 relevant family of states, the metric, the tolerance, the communication graph
469 and the deadline are all declared inputs, as are the represented variables and
470 the fidelity counted as accurate. Reading the map as temporal neural pre-
471 diction would require dynamics; reading the resulting code count as physical
472 memory capacity would require a hardware model, which a state-dependent
473 readout would have to extend.

474 **6 What the conditions constrain in a digital** 475 **candidate**

476 The results of Section 2 hold for any candidate, and for a large language
477 model (LLM) running on graphics processing units (GPUs) four of them are
478 checkable rather than merely stateable. Those four are where this framework
479 bears on claims made for such systems. It does not exclude them, and no re-

480 sult here could; what it does is fix what a claim of unity or self-representation
481 commits its maker to measuring.

482 **Codes, not reports.** Reconstruction across a family is a counting con-
483 straint before it is anything else. If two macrostates the candidate is said to
484 represent satisfy $E(s) = E(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out recon-
485 structing both within ε , whatever the readout. A system’s reports are read-
486 outs. Identical verbal reports across two conditions are therefore consistent
487 with identical internal codes and settle nothing about the states, while the
488 converse claim — that the system represents its own condition across a de-
489 clared family — asserts that its internal code separates that family, which is a
490 statement about activations under intervention and not about text. Naming
491 the channel makes the count concrete. Where the code is the token sequence
492 a model emits—which is what its next step reads—the alphabet is the de-
493 ployed vocabulary (`tokenChannel`), and a family reconstructed through it
494 is no larger than the vocabulary size raised to the number of steps (`card_`
495 `le_card_tokens`). Over an extended exchange that number is large and the
496 bound is slack; at a single step it is one vocabulary’s worth, so a claim that
497 one emission reports the model’s own condition across a declared family as-
498 serts that the vocabulary separates that family (`card_le_card_tokens_one`).
499 A claim resting on activations rather than on text has declared a different
500 channel, to which the same count applies with that channel’s alphabet. The
501 measurement follows from the statement: declare the family and the fidelity

502 first, then test whether the encoding region distinguishes its members.

503 **The region, not the readout.** Where the encoding is read as the state's
504 restriction to a region of the machine, the demand falls on the region before
505 any readout exists. For a decoder-only transformer the causal mask makes
506 the network an instance of the update model of Section 5: the activation at
507 position t and layer ℓ reads positions at most t at layer $\ell - 1$ and nothing
508 else, so a region's causal past is fixed by the architecture rather than chosen
509 by interpretation. That instance is formal, and what it establishes is sharper
510 than slow growth: the causal past of position t stays inside the positions
511 at most t at every depth (`mask_ball_subset_le`), so the deadline bound of
512 Section 5 fires at every deadline at once rather than at a chosen one, with
513 its communication graph given rather than assumed (`mask_no_guarantee`).
514 That is a statement about one forward pass; across steps a model does read
515 its own prior output, which is the channel the preceding paragraph counts.
516 Read against the coupling proposed here the asymmetry runs both ways, and
517 neither way is an advantage. A kernel of full support has no neighbourhood
518 structure for the bound to bite on—after one round the causal past is the
519 whole substrate, so the hypothesis that fires the obstruction is unsatisfiable at
520 any deadline but zero (`not_outside_past_of_isFullSupport`)—but a physi-
521 cal field propagates at a finite speed, so treating one that way is a calibration
522 against the phase time scale and not a theorem. What the comparison does
523 fix is where the bound has force. Not on the attention graph, where a position

524 reads every position at or before it and the fan-in is the context length. On
 525 the physical interconnect, where fan-in is genuinely bounded: a graph whose
 526 sites each hear from at most ν others reaches at most $(T + 1)\nu^T$ sites in
 527 T rounds (`card_ball_le_mul_pow`), so a model sharded across many devices
 528 pays a number of rounds growing like $\log_\nu$ of the device count for guaranteed
 529 agreement, and buying that deadline back costs wiring—reaching every site
 530 at a single round requires a device hearing from all but one of the others
 531 (`le_degree_of_reaches_one`). Both ν and the device count are declared, as
 532 the communication graph and the deadline already are, and what the count
 533 bounds is the graph a guarantee requires rather than what hardware is build-
 534 able. A claim that a model reports on its own processing consequently carries
 535 an obligation to say which positions and layers the report’s causal past con-
 536 tains, and a report that appears to track an intervention outside it indicates
 537 a confound or a different region, not a faster channel.

538 **Agreement about what.** Unity requires the candidate’s local descriptions
 539 to agree on independently identified shared quantities (Section 4). Common
 540 timing, a shared residual stream and fluent output supply none of that, for
 541 the reason Section 2 gives: coordination bounds content disagreement only
 542 through encoders already related on overlaps, and which quantities two parts
 543 of a model describe in common is a decoding question carrying the failure
 544 modes of Section 8. The cover is a choice, and choosing it to secure agreement
 545 empties the claim.

What is not excluded. Digital representations can satisfy compatibility,
 so finite hardware is compatible with the gluing construction and none of
 the above is a no-go for machines. The coupling-gated route E56–E78 is a
 different matter: it requires a calibrated physical coupling, its noise regime
 and a connection from coordination to content agreement, and an attention
 matrix or a shared clock supplies none of these identifications. Computing a
 field model likewise leaves its realization in the computing substrate to be es-
 tablished. Those physical obligations fall on the cortical proposal identically.
 The capacity comparison those support comparisons lack is available, and it
 favours neither side by construction. A continuum code space read at resolu-
 tion δ holds at least its volume divided by the volume of a resolution cell, so
 it out-resolves an alphabet of 2^b codes exactly when that volume exceeds 2^b
 cells (`two_pow_lt_packingNumber_of_lt_measure`); below that threshold the
 inequality runs the other way, and at a coarse enough resolution the region
 holds a single code. What decides the comparison is the volume, the res-
 olution and the dimension, all measured quantities, and continuity decides
 nothing on its own: a continuum code space read at a finite resolution is
 a finite alphabet whose capacity is a bit count. For the same reason this
 account credits an analog medium with no computation beyond a Turing
 machine’s. Real-valued states carry unbounded capacity only at unbounded
 precision, and the noise floor $D > 0$ that Eq. (1) is stated against removes
 it.

The installed-energy condition of Section 3.2 is restricted in the same

569 way. It constrains a macroscopic candidate only through κ , which is hard-
570 ware data supplied by mode profiles and prices, and Supplemental Material
571 reads the condition against a cortical metabolic budget to return the conver-
572 sion factor it would require rather than a verdict. The hardware comparisons
573 there have similarly restricted scope: fixed wiring can impose a correlation
574 disadvantage, and point-supported kernels vanish under an atomless contin-
575 uum measure, but neither comparison makes the representation conditions
576 unattainable on GPUs. A universal exclusion would need a necessary condi-
577 tion for consciousness and a proof that every relevant implementation violates
578 it, and the present composition supplies neither.

579 Whether experience requires a living substrate is the explicit subject of
580 biological naturalism (Seth, 2026a,b). On that view the properties ground-
581 ing experience are those of self-sustaining organisms, and simulating them
582 does not secure them. The conditions stated here name no biological mech-
583 anism, but computation described as such does not satisfy them either: a
584 candidate must exhibit a physically realized coupling in a measured noise
585 regime, agreement about independently identified shared quantities, and an
586 internal reconstruction meeting a declared fidelity. Whether a realization
587 of those conditions that is not alive would be accompanied by experience
588 is the point at issue, and these results do not settle it. A related position
589 locates the difference in the mode of computation rather than in life as such:
590 biological systems compute across scales that do not separate, and along-
591 side discrete operations carry continuous-valued dynamics belonging to their

592 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
593 here is one measurable instance of that requirement, fixed to a physical field
594 in a declared noise regime, and a description of such a field does not satisfy
595 it.

596 The neuroscientific case against *human-like* consciousness in text-only
597 LLMs concerns embodied sensory interaction, thalamocortical organization
598 and biological regulation linking perception and action to survival (Aru et al.,
599 2023). Their absence supports skepticism, conditional on their causal rele-
600 vance to experience. Those mechanisms are not requirements of the abstract
601 gluing and reconstruction definitions, and workspace-like representations re-
602 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to
603 investigate. What this framework adds to that investigation is an order for
604 the questions: declare the family of states and the fidelity, fix the encoding
605 region and read its causal past off the architecture, and only then ask what
606 the system reconstructs. A claim that survives those three steps is a claim
607 about the machine; one that does not survive them was a claim about its
608 output.

609 7 Thermodynamics and composition

610 The phase and representation results concern coupling, content disagreement
611 and reconstruction error. Thermodynamics constrains the physical opera-
612 tions that could implement them, and it does so from the side rather than

613 along the spine: no edge below derives coherence, compatibility or reconstruc-
 614 tion from a heat bound, and the resource condition these edges terminate in
 615 is necessary without being restrictive for any macroscopic candidate (Sec-
 616 tion 3.2). What this half contributes is a ledger in which a resource claim
 617 is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$;
 618 erasure heat depends on specified system–bath assumptions (Landauer, 1961;
 619 Norton, 2005; Shenker, 2000). A separate boundary result concerns a field
 620 reaching disconnected vacuum components (Kibble, 1976). The composition
 621 connects these objects through the explicit hypotheses in Table 1.

622 7.1 Feedback changes the information constraint

623 For passive memory X_t driven by an environmental signal S_t , the predictive
 624 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

625 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 626 et al., 2012). Zero retained information also satisfies it, so a task and a
 627 learning rule require additional specification. Relating this bound to Lan-
 628 dauer heat requires a common thermal scale and budget, and memory waste
 629 bounded by erased entropy.

630 An acting agent changes the signal it will receive. Specify policy, world,

631 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
 I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
 &\quad - I(X_t; S_{t+1} \mid S_t).
 \end{aligned}
 \tag{12}$$

632 The second conditional term vanishes in the passive model. With feedback,
 633 the signed difference can be negative as action creates correlations with the
 634 future. A causal reading requires an environmental state sufficient to describe
 635 the relevant dynamics.

636 For a controlled substep, the active thermodynamic bound instead uses
 637 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
 638 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with thermal scale
 639 $k_B T > 0$ and heat positive into the reservoir (Horowitz and Esposito, 2014;
 640 Ito and Sagawa, 2013). Every forward path of positive probability must have
 641 a positive-probability reverse for this finite identity. A missing reverse gives
 642 infinite divergence. The supplement develops the support conditions and
 643 composes actuation with memory updates on their shared intermediate law.

644 **7.2 Learning and repeated operation need explicit re-** 645 **sources**

646 The finite examples distinguish selecting a feasible policy, relaxing a register
 647 toward an encoded reward, and learning an unknown task from observa-
 648 tions. The observational learner acquires correlation with an environmental

parameter while its action marginal stays uniform. A continuing model executes actuation, observation-driven learning and reset in sequence, preserving above-chance performance at a positive work cost per cycle.

A phase-derived sensor makes the informational distinction concrete: an informative readout and a blind control have the same phase order but different learning performance. Attaching the learner’s register to the actuator of Section 3.2 ties reward, installed kernel and work to its own executed law. The learning witness rises above the necessary installed-energy bound while its matched control remains on the exclusion line. These finite constructions, detailed in Supplemental Material, leave the evolving feedback between phase dynamics, installation and sensing to be specified.

For that model, common energy observable $\mathcal{E}$, thermal scale $k_B T$, local detailed balance and positive support at each stage give

$$k_B T [H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N \leq B + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N, \quad (13)$$

where expected cumulative work stays within the supplied allowance B at every prefix. This bounds expectations. Separate constructions track a store on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles,

while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 3.2 requires physical relations that these separate models do not supply automatically.

7.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them. The regime in which those premises are exhibited is a near-equilibrium one: strictly positive forward and reverse elementary paths, local detailed balance at one thermal scale, and entropy differences and heats of order $k_B T$, the witnesses being two-bit systems with interaction energies of that size. Cortex is not in it. The installed energy of Section 3.2 is 2.4×10^{14} times kT at body temperature over one region and one phase-diffusion time, and a budget B

688 of that size stands against an entropy term of order $k_B T$, so Eq. (14) is
 689 satisfied by any macroscopic candidate and excludes none of them. Reading
 690 the register instead at the scale where its premises are physical, such as a
 691 single channel, relocates the difficulty to E45Active, which would then have to
 692 carry one molecular step’s budget to the convergence of a continuum kernel.
 693 The active branch is an alternative route to the same node rather than a
 694 stronger one, and no result of Section 3.2 passes through it: the installed-
 695 energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.

Table 1 (continued)

Connection	Status	Required assumption
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .

Table 1 (continued)

Connection	Status	Required assumption
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact overlap compatibility and a reflexive boundary specifying encoding and readout. They conclude **UnifiedSelf**: the glued section is the unique fixed point of self-prediction. Compatibility and accuracy at that section enter as premises; uniqueness follows. Identifying encoding with restriction needs the separate resonance condition of Eq. (8).

Each of the eight edges is supplied rather than derived from the one above it. E12 is equivalent to its conclusion, so capacity and boundary are independent roots rather than a step from one to the other; both E45 discharges take their convergence from refinement data and not from the thermodynamic bound; E78 states that coherence relaxes the supplied cover, and the coherence-to-content estimate of Eq. (6) bounds disagreement only for encoders already shared on overlaps. The inferential content of the composition is its last step, where contraction and a fixed glued section give uniqueness. What composing them establishes is that the premises are consistent, named

711 in one place and machine-checked together, and not that coherence follows
712 from installed energy or self-representation from coherence.

713 Joint witnesses combine a register, boundary, refining mesh and small
714 spatial cover; active witnesses add controlled thermal updates and a priced
715 installation. They establish satisfiability under their stated identifications.
716 Mesh convergence follows from refinement data, and the witnesses establish
717 no common cortical mechanism. Supplemental Material maps the claims to
718 their Lean declarations and records the counterexamples. The development
719 declares no additional axioms beyond its libraries' foundational axioms.

720 8 What the observations establish

721 The composition holds under its premises. Whether observation can estab-
722 lish those premises is a separate question, and the three requirements it
723 connects concern different objects: phase dynamics, local descriptions and
724 internal encodings. Their observables must retain these distinctions. The
725 following controls show which mechanisms remain indistinguishable, which
726 temporal approximations fail, which states one observable cannot separate,
727 and what independent content measurements must establish. Each is a finite
728 numerical regime on a declared model, with full parameters and controls in
729 Supplemental Material, and none of them calibrates a cortical quantity.

730 8.1 Stationary distributions leave the mechanism un- 731 determined

732 Equation (2) gives a concentration–coherence relation with no fitted shape
733 parameter. Estimate r from circular resultant length and a from density
734 shape: inverting R from the measured r would make agreement automatic.
735 Even independently estimated agreement identifies a statistical family shared
736 by different mechanisms.

737 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
738 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
739 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
740 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
741 and diffusion D_0 have that same density. Their stationary distributions co-
742 incide window by window despite different causes.

743 Resolved dynamics supply additional information. In the ideal model with
744 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
745 increment means constrain drift. Separating endogenous coupling from a
746 matched common drive additionally requires a measured drive or controlled
747 perturbation; identifying electromagnetic coupling requires evidence specific
748 to that interaction. Supplemental Material gives the equivalence construction
749 and its numerical check.

750 8.2 Finite ramps change the observable onset

751 The square-root law in Eq. (3) describes stationary solutions. A changing
752 coupling requires time for the phase distribution to approach those solutions.
753 In a deterministic mean-field control that crosses the threshold symmetrically,
754 none of the scanned rates meets the declared tracking tolerance: slowing
755 the crossing does not remove the terminal residual under that initialization.
756 A slow control variable therefore supplies no general guarantee of branch
757 tracking at onset. Supplemental Material gives the residual, rate range and
758 admissible rates away from threshold.

759 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
760 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
761 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
762 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The
763 6 uncensored legs supply the delay scaling below; the decade-spaced ones
764 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
765 onset exponents differ from those of the exact stationary branch evaluated
766 on the same grids. At the fastest speed the ensemble mean never reaches the
767 fit window, so its fitted slope does not estimate stationary onset.

768 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
769 makes the apparent exponent depend on ramp speed and fitting window.
770 The specified deviation-based critical ramp speed is not bracketed by this
771 sweep. With fluctuating coupling, the relevant stationary comparison also
772 depends on rate: the mean coupling approximates order in the tested fast

773 regime, while averaging stationary order over coupling works for the tested
774 slow drives confined to one side of threshold. Slow crossing drives can satisfy
775 neither approximation.

776 **The prediction.** The delay scaling is what this section supports as a pre-
777 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
778 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of $[0.394, 0.494]$,
779 which does not contain $1/2$. What that shortfall measures is the escape cri-
780 terion and not the scaling. Applied to a deterministic mean-field run seeded
781 at the fluctuation floor, the same estimator returns 0.425 over the same legs
782 at the same escape criterion, inside the interval on the measurement. That
783 run has no noise to delay it, so the shortfall is the criterion's: it crosses the
784 saturation between the floor it starts from and the level it is stopped at, and
785 tightening the criterion to $r \geq 0.05$ returns 0.490. The bias is reproduced
786 where the answer is known, which is what licenses reading the measurement
787 as the law rather than as a competing exponent.

788 Stated as a law rather than as a number: onset delay scales as the square
789 root of the rate at which coupling crosses threshold. Testing it requires
790 only that $\dot{K}$ be monotonic in a manipulated rate — not its value, nor the
791 installation factor κ , nor the field-to-coupling conversion rate γ of Eq. (15)
792 — so it is a within-subject design that varies the rate of emergence from
793 anaesthesia alone, with each participant as their own control. Such a design
794 must avoid the hazard this sweep exhibits: a fixed absolute escape criterion

795 meets a size-dependent critical fluctuation floor, so the criterion has to be
 796 referred to each recording's own pre-threshold fluctuation scale rather than
 797 to a shared number. The diffusion D is the remaining input, and Section 8.1
 798 notes that it is estimable from resolved phase traces through the rate $2D$
 799 at which quadratic variation accumulates, which is a different quantity from
 800 the concentration the propofol exercise could not range over.

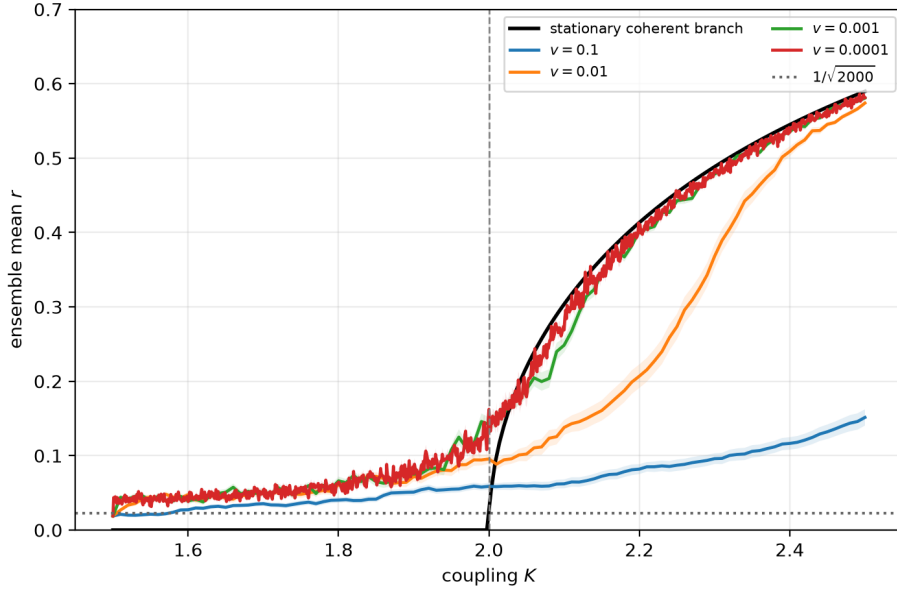


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

801 Additional controls address integration bias, population size and spatial
 802 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 803 $[1.958, 2.009]$ contains 2.

8.3 A winding state leaves the order parameter ambiguous

The order parameter r is indifferent to where a phase sits in space. A field winding once across a periodic sheet therefore has $r = 0$ while every patch of it stays locked. Such a field is not an artificial construction here. On a torus a winding is stationary for any isotropic symmetric kernel at identical frequencies: the coupling term is odd in the separation against a kernel even in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves this for any finite periodic array, this sheet included. A winding satisfies the positivity and identical-frequency hypotheses. It violates only the quarter-turn initial spread that the convergence result additionally requires before placing the limit at the uniform state.

On the sheet of Section 3's spatial control, a winding survives below an interpolated decay length of 0.1871 mm and unwinds above it (Fig. 4). That boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the same coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start and $r = 0.0083$ from a winding one. Both are stationary, with patch-local order 0.9015 and no winding defect. At fixed K the observable therefore takes two values spanning nearly its whole range. Coupling does not determine the resultant, and a low resultant does not by itself distinguish weak coupling from a winding field.

Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-

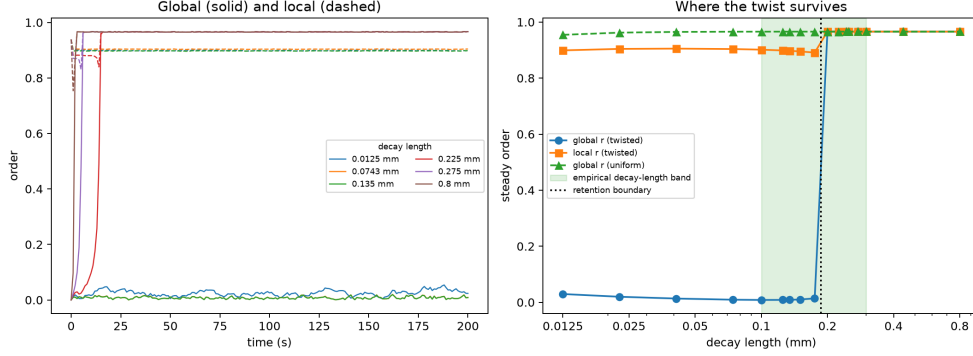


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

826 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 827 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
 828 Heterogeneity shortens the range that sustains a wave without removing it.

829 Which windings persist has a criterion behind it. Linearizing the phase
 830 model about a winding returns a coupling of the same translation-invariant
 831 form, each separation reweighted by the cosine of the phase the winding
 832 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
 833 again even in the separation, so the same characters diagonalize it and ev-
 834 ery mode’s rate is one of its eigenvalues. Where the winding advances by
 835 at most a quarter turn across every separation the coupling reaches, the

836 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 837 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 838 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 839 longer separations is what carries a phase advance past a quarter turn, and
 840 that is the mechanism the decay-length sweep varies. The criterion bounds
 841 rates rather than trajectories: asymptotic stability is not proved, so the
 842 states that persist occupy a narrower range than it admits, and the bound-
 843 ary quoted above is measured and not derived from it. The same kernel is the
 844 potential's curvature, so a winding at $r = 0$ is not the global minimum the
 845 convergence result places its limit at, and one carrying a growing mode is not
 846 a local minimum either: that result's quarter-turn hypothesis must exclude
 847 a winding rather than merely decline to cover it (`char_not_potential_min`).

848 An amplitude layer decides which of these states exist, and leaves what
 849 the resultant reports unchanged. Every oscillator of the phase model sits on
 850 the unit circle, so a field with an amplitude node has no representation in
 851 those dynamics. The real Stuart–Landau extension supplies one on the same
 852 coupling, and leaves intact the gradient structure the convergence results rest
 853 on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\Gamma - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its
 854 phase equation is the Kuramoto equation with each coupling weighted by a
 855 ratio of amplitudes (`amplitude_phase_is_kuramoto`). The phase model is
 856 this one with the amplitude ratios set to one, rather than a separate object.

857 What the added degree of freedom decides is which windings exist at all.
 858 On a periodic array of any dimension, this sheet included, a winding requires

859 $\Gamma + \lambda > 0$, with Γ the layer's linear gain. Here $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ is
 860 the eigenvalue the kernel supplies, summed over separations, with $\Delta\theta_d$ the
 861 phase the winding advances across d (`exists_charState_iff`; on the ring
 862 $\Delta\theta_d = 2\pi qd/n$). For the nonnegative kernels used here $\lambda \leq 0$, and moving
 863 coupling mass to longer separations lowers it further. A broad enough kernel
 864 therefore removes the winding outright. The sweep above reaches that same
 865 boundary by a different route: the phase model retains every winding at
 866 every coupling, and loses this one under integration instead. The two layers
 867 bound the same regime from opposite sides. A phase recording does not
 868 separate a state the amplitude layer excludes from one the dynamics fails to
 869 sustain.

870 Three limits bound the layer for a reader of the cortical literature. These
 871 states are windings and not spirals, and on the sheet that is a theorem rather
 872 than a shortage of dimensions: equal modulus at every site puts the phase
 873 circulation at exactly zero on every plaquette, and on a periodic sheet the
 874 circulations of *any* field sum to zero, so a lone defect is not a state of this
 875 geometry at all. Relaxational dynamics has no propagation speed, which is
 876 the quantity a cortical wave measurement leads with. And $|z|$ is a normal-
 877 form radius, which a measured envelope is not; this work identifies the two
 878 nowhere. Supplemental Material gives the sweep, its uniform control, the
 879 amplitude layer, the standing wave it admits inside the band, the potential's
 880 curvature at a winding and the defect theorems, together with the remaining
 881 limits of both.

882 A vanishing resultant is the absence of global agreement and not of agree-
 883 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrep-
 884 ancy bound of $2\sqrt{2}L|\sin \Delta\theta|$, with $\Delta\theta$ the phase advanced across the patch,
 885 which falls to zero as the same winding is spread over more sites (`chord_le-`
 886 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
 887 Encoders shared on overlaps therefore still bound content disagreement in-
 888 side a patch of a state that no global observable calls coherent. The winding
 889 number itself stays unconnected to content. It is, however, a property of the
 890 state rather than of the cover read on it. On a ring the whole turns dropped
 891 by the winding's affine local lifts are zero at every site but the wrap, where
 892 the residue resets, and they sum around the loop to q ; a nonzero q obstructs
 893 every global real-valued phase lift, so one state at one coupling has resultant
 894 exactly zero and degree q (`winding_degree_obstructs`). Reweighting the
 895 sites moves the first quantity and redrawing the loop's intermediate transi-
 896 tions leaves the second where it was. The scope is the ring's: this is a loop's
 897 winding and not a spiral, the transitions are those of the affine lifts rather
 898 than of an unwrapped measurement, and the integer is identified with no
 899 cortical observable.

900 This qualifies what a phase trace can establish. An onset criterion stated
 901 in r alone registers a change of basin as readily as a change of coupling.
 902 The protocol of Section 9.3 therefore needs a spatial observable beside the
 903 resultant.

904 8.4 Coherent adaptation can lose environmental align- 905 ment

906 An adaptive coupling rule might supply representational structure, but its
907 objective must reward the relevant agreement. For symmetric coupling, pair-
908 wise sine terms cancel in the population mean drift. Minimizing squared drift
909 at that fixed mean therefore rewards equalizing drift across nodes. In the
910 frequency-cluster control, this favors coupling between unlike-frequency clus-
911 ters, whereas the diagnostic environmental template rewards coupling within
912 clusters.

913 The formalization proves descent of the squared-drift objective for a spec-
914 ified gradient flow at fixed phases. Its functional differs from thermodynamic
915 entropy production: at a coherent stationary density the probability current
916 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
917 rial gives the operator, its gradient and the conditions of the descent result.

918 The finite control co-evolves noisy phases and symmetric, nonnegative
919 couplings at fixed total resource. Its template is a diagnostic covariance of
920 latent signals, never a training input. Across three seeds, second-half order
921 remains between 0.9167 and 0.9222, while the within-cluster to between-
922 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5).
923 Random-update, permuted-gradient and frozen-kernel controls are reported
924 in Supplemental Material.

925 The immediate post-update objective reduction disappears when aver-

926 aged over complete update intervals. A prespecified grid of 6 rate–cadence
927 combinations contains no case meeting the 5% sustained-reduction criterion
928 while retaining coherence and losing alignment. The finding is coherent adap-
929 tation away from the specified structure, with no demonstrated sustained cost
930 reduction.

931 Kernel alignment and overlap compatibility are different observables.
932 This control addresses acquisition of its designated template under its par-
933 ticular objective. Learning models can explicitly include data in an objective
934 (Corberi et al., 2026); an account of agreement about the cup must specify
935 the content its learning rule is meant to preserve.

936 8.5 Content agreement needs independent decoders

937 A neural test needs independently identified domains, local descriptions and
938 restriction maps. Two decoders should estimate a shared content on over-
939 lapping domains before their agreement is examined. Discrepancy requires
940 a null under decoding noise and controls using deliberately compatible and
941 incompatible examples. An implementation on constructed data separates
942 these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by
943 trial; the compatible construction scores 0.476 on an interval containing one
944 half. Shared-prior shrinkage can nevertheless manufacture agreement, and
945 regional bias can manufacture disagreement. Per-patch decoding diagnostics
946 detect these failures; the compatibility statistic alone does not.

947 Phase agreement cannot replace that measurement. On constructed con-

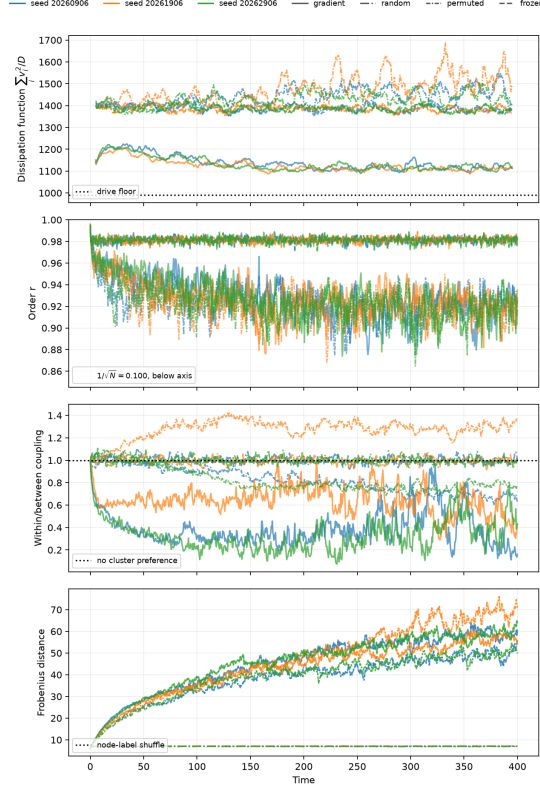


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

948 tent, phase carries a share 1.000 of the content entropy when content is built
949 from it and 0.000 when it is not. A phase-derived statistic reports com-
950 patibility on phase-locked patches whose local descriptions disagree. The
951 implemented content statistic detects their incompatibility. These controls
952 validate a measurement on known constructions; a neural content model and
953 decoder remain to be established.

954 Choosing the cover also chooses a candidate subject: trivial descriptions
955 can agree while carrying little content, and a partition can conceal disagree-
956 ment. The useful choice must therefore be constrained by what neural ac-
957 tivity encodes. The exact composition assumes compatibility of its supplied
958 cover. The content mechanisms above identify additional hypotheses under
959 which agreement can arise; they leave the empirical choice of content and
960 the connection to that cover explicit.

961 **8.6 What an informative phase-density measurement** 962 **requires**

963 The concentration range and the dependence structure decide whether the
964 concentration-coherence relation can be tested at all. Under the specified
965 log-density estimator and bias-floor criterion, discrimination at the observed
966 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
967 designs, rising to 31000 when sites share a mean in groups of 10. These
968 counts describe that estimator and dependence model; the intended recording

969 requires its own calibration. Supplemental Material reports the bin-count
970 controls and the full dependence-aware design.

971 That ceiling comes from exercising the estimators on public propofol-
972 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows.
973 Bipolar and common-reference montages produce different residuals. The
974 bipolar results are compatible with the Bessel relation over a concentration
975 range too narrow to distinguish it from its linear approximation at the ob-
976 served scatter, which is the range the design scan takes as given. Temporal
977 pooling changes the estimand, and sedation recordings leave the awakening
978 hypothesis untested.

979 An informative dataset therefore needs sufficient concentration range or
980 precision, calibrated estimators and an observation model for dependence,
981 source mixing and reference choice. Public mouse calcium imaging provides
982 thousands of spatially located neurons across waking, sleep and anesthe-
983 sia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
984 estimation band used here. Site count, state coverage and bandwidth are
985 separate requirements.

986 **9 The cortical hypothesis and an awakening** 987 **test**

988 The cortical proposal identifies a physical source of coupling and asks whether
989 its changes help explain recovery. The tests in Section 8 determine what

990 observations must distinguish before a phase trace can support this identifi-
991 cation.

992 **9.1 Physical identification and calibration**

993 The cortical hypothesis identifies part of effective coupling with endogenous
994 extracellular electromagnetic interactions. Fields can modulate spike timing
995 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
996 propose coordination of distributed neural information (Pinotsis & Miller,
997 2022; McFadden, 2020). Phase organization under propofol motivates study-
998 ing this regime (Bastos et al., 2021), but does not measure coupling.

999 The formal field predicate requires a jointly continuous kernel, a normal-
1000 ized substrate, coupling equal to the kernel average, and no single site pro-
1001 viding more than half that strength. Toy constructions satisfy it. A cortical
1002 realization must specify the scale at which the kernel and its reduction ap-
1003 ply: calcium imaging reports different state-dependent modularity patterns
1004 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

1005 Cortical columns are the obvious candidate unit, and they answer one
1006 of the two demands the formalism makes. As the oscillator index they are
1007 a serious proposal: they are delineated by response properties and anatomy
1008 rather than by the agreement they would be used to establish (Mountcastle,
1009 1997), and a phase can be estimated from a column-sized territory at the
1010 electrode spacings used for cortical wave measurements (Townsend et al.,
1011 2015). As the content cover they are a poor one, for three separate reasons.

1012 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
1013 the compatibility obligation without constraining anything (Section 4). The
1014 cover condition additionally requires the patches to exhaust the substrate,
1015 while much of association cortex has no accepted columnar parcellation. And
1016 the column's standing as a canonical cortical unit is contested (Horton &
1017 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
1018 supplies the oscillator index and leaves the content cover to be identified on
1019 separate evidence, which is the bridge assumption E78 records.

1020 The spatial reduction is also testable. A row-sum aggregation at zero
1021 diagonal tracks the uniform-coupling control over the tested spatial decay
1022 range, while frequency heterogeneity displaces the operational threshold. The
1023 fit window biases both estimates relative to $2D$; agreement with the control
1024 is therefore the supported conclusion. Supplemental Material reports the
1025 reduction, its numerical error and the distinct assumptions needed for the
1026 kernel-average identification in Eq. (5).

1027 The field commitment requires physical evidence beyond a choice of math-
1028 ematical representation, and the available support comparisons do not supply
1029 it. A finite matrix spread over positive-mass cells preserves its total cou-
1030 pling and phase correlation exactly, while point-supported vanishing under
1031 an atomless measure and a disadvantage from excluding a better-correlated
1032 pair concern specified support constraints; Section 6 reads the same compar-
1033 isons against a digital candidate. Neither supplies energy prices.

1034 A spike-time shift alone supplies no coupling rate. The dimensionless

1035 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
 1036 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (15)$$

1037 This rate describes how induced timing shifts accumulate into drift. Nei-
 1038 ther it nor the installation factor κ follows from a heat bound. Testing the
 1039 physical threshold requires independent estimates of the interaction kernel,
 1040 its aggregation and diffusion in the same cortical regime.

1041 **9.2 Extracellular geometry and recovery**

1042 The proposed control variable is extracellular geometry. Extracellular vol-
 1043 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
 1044 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
 1045 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
 1046 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 1047 an effective-medium description predicts that a larger extracellular fraction
 1048 raises conductivity and reduces field amplitude.

1049 The coupling-gated hypothesis consequently predicts that extracellular
 1050 contraction on awakening raises E and, holding the other factors in Eq. (15)
 1051 fixed, raises effective coupling. This argument supplies a direction; the
 1052 geometry–coupling relation, its magnitude and its role in human recovery
 1053 require measurement.

1054 For a smooth crossing at nonzero speed, evaluating the stationary branch
 1055 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

1056 This follows from Eq. (3) at fixed D and concerns the stationary branch
 1057 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 1058 dently justified tracking approximation. The crossing controls in Section 8.2
 1059 leave that approximation unsupported at onset, even though a separation of
 1060 geometry and phase time scales can support it away from threshold. The
 1061 awakening hypothesis therefore needs a calibrated dynamical prediction for
 1062 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
 1063 does not supply a recovery latency.

1064 **9.3 A protocol for the awakening experiment**

1065 The physical test has six requirements:

- 1066 **1. Coupling and diffusion in the same units.** Estimate them from re-
 1067 solved phase dynamics, separating heterogeneity and observation noise.
 1068 Measure the drive or perturb the system to distinguish endogenous cou-
 1069 pling.
- 1070 **2. Geometry calibrated to coupling.** Validate the geometry observ-
 1071 able and its relation to effective coupling, with uncertainty. The con-

ductivity argument supplies only the expected direction.

3. **Calibrated phase measurements.** Estimate r by resultant length and a by unweighted regression of log density on centered cosine. Choose site count and binning from a dependence-aware calibration over the intended concentration range. The tested designs require thousands of sites at the observed concentrations; temporal pooling changes the spatial estimand.

4. **An independent recovery endpoint.** Measure behavioural recovery separately from phase organization.

5. **A predeclared dynamical onset criterion.** Set a coherence threshold above the finite-size floor and evaluate its latency under the measured coupling trajectory, including ramp speed or fluctuations. Record a spatial observable beside the resultant, since a winding field reports a low resultant at coupling that sustains local order (Section 8.3).

6. **Matched model comparison.** Compare coupling increase, diffusion change, common drive and alternative recovery models under the same observation and filtering assumptions.

Failure of the concentration–coherence relation over an informative range constrains the density model. Failure of the calibrated geometry–coupling association constrains the physical gate. Better prediction of independent recovery by a competing mechanism counts against its proposed role. A

1093 favorable outcome still leaves the content of coordinated activity to be ex-
1094 plained.

1095 **10 Discussion**

1096 Two things have been established here, and they are of different kinds. The
1097 constraints of Section 2 hold outright: synchrony does not deliver compatible
1098 content, and where shared encoders let it bound content at all, the bound
1099 that holds uniformly over pairs does so only inside patches too small to
1100 span the territories at issue, distance costing either a connected cover or
1101 a locked configuration and a declared spectral gap; the resultant does not
1102 determine the state; a unique fixed point is not a reconstruction; a region that
1103 cannot resolve a family cannot encode it; and agreement about a change takes
1104 communication. Any account that reads unity off phase coherence has to
1105 meet them, this one included. The account set out in Section 1 is conditional
1106 at every step by contrast: each connection in it is a hypothesis supplied
1107 rather than a result derived, and Table 1 names all eight. Its four non-
1108 standard ingredients, named in Section 1, are the installed-energy condition,
1109 the patch-local form of the content bound, the criterion separating a winding
1110 field from weak coupling, and the machine-checked composition with its joint
1111 witness. What follows takes the conditions in turn, as commitments the
1112 proposal incurs, as measurements that would bear on them, and as questions
1113 the framework leaves open.

1114 10.1 Unity and a minimal self as empirical commit- 1115 ments

1116 We propose the state glued from compatible local descriptions as a corre-
1117 late of conscious unity, and its accurate internal reconstruction as a correlate
1118 of a minimal self. The motivations are that different contents belong to
1119 one experienced situation and that the situation is experienced from a per-
1120 spective. The conditional results identify relations a realization would need:
1121 sufficient installed resources for the specified phase model, agreement about
1122 shared content, and an encoding that supports reconstruction at the required
1123 fidelity. Discharging E12–E89 for cortex would make the composition a state-
1124 ment about cortex: that a cortical field above threshold relaxes to a cover
1125 whose compatible local descriptions glue, and that the glued state is the
1126 unique fixed point of a reconstruction map realized there. It would leave the
1127 two identifications above exactly where they are, because neither is among
1128 the eight and neither follows from their conjunction; they are the commit-
1129 ment this subsection names, and Section 10.2 lists the three observations
1130 that would count against it.

1131 Compatibility allows different descriptions to contribute different infor-
1132 mation. It also permits uncertainty and conflicting judgments: apparent cup
1133 position and judged cup position can be distinct quantities in a consistent
1134 description of their discrepancy. Which quantities are shared must be de-
1135 termined independently. Choosing them solely to secure agreement would

1136 deprive the proposal of empirical force.

1137 The minimal-self proposal concerns the perceiver’s position and condition
1138 within the situation, such as the cup’s location relative to a body that can
1139 reach it. The encoding could be implicit and need carry no autobiograph-
1140 ical narrative. Its proposed necessity is philosophical: even an unreflective
1141 episode has a perspective. Gluing by itself requires no self-encoding, and re-
1142 constructability alone supplies no functional role. Altering the encoded hand
1143 or viewpoint while holding the external scene fixed should change estimates
1144 of the perceiver’s relation to the cup as the readout predicts. Whether that
1145 represented perspective is experienced remains a further question.

1146 **10.2 What a test of the full proposal must establish**

1147 Both uniqueness results concern specified objects. They choose neither the
1148 cover nor the encoding, and do not select between candidate subjects. In-
1149 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
1150 here selection requires independent evidence about neural organization. A
1151 constant readout can contract and fix a predetermined state while discard-
1152 ing its input. The family criterion in Section 5.2 tests the sensitivity that
1153 this construction lacks, provided the relevant states and fidelity are declared
1154 independently.

1155 A test must separately measure coupling relative to diffusion, content
1156 compatibility and self-reconstruction, alongside an independent endpoint for
1157 experience. Meeting all three without a unified episode would challenge suffi-

1158 ciency. A unified episode with incompatible descriptions would challenge the
1159 unity identification; one lacking the specified self-encoding would challenge
1160 the reflexive requirement. Awakening, dreaming and dissociative states offer
1161 possible settings, but absence of report cannot by itself define absence of
1162 experience.

1163 The next physical requirement is a calibrated geometry–coupling relation
1164 and resolved phase dynamics. The installation bound contributes a nec-
1165 essary resource condition where its hardware representation applies; cortical
1166 mode prices and field-mediated coupling require measurement. For the repre-
1167 sentational connection, an informative experiment asks whether a controlled
1168 change in coordination alters independently decoded content agreement, with
1169 controls for sensory information and decoder quality. The constructed-data
1170 estimator establishes feasibility for specified contents. Neural local descrip-
1171 tions, shared overlap encoders and the self-encoding have yet to be identi-
1172 fied. The population-size dependence of the phase-to-content bound limits
1173 its quantitative use in the form Section 4.2 reads off the sheet, and each
1174 way around it asks the measurement for something further: a located excep-
1175 tional set, before a count of disagreeing pairs bears on a named overlap; a
1176 cover whose nerve is connected and whose patch diameters are measured; or
1177 a frequency-locked population over a coupling graph whose connectivity is
1178 measured structurally rather than estimated from the activity the bound is
1179 about.

1180 The separation results explain why these measurements must be inde-

pendent. A stationary distribution leaves several recovery mechanisms indistinguishable. Equal phase order can accompany incompatible descriptions or different task information. The tested plasticity rule preserves coherence while losing its designated template. Each finding concerns the specified model and observable; their role is to delimit the evidence needed for the proposed connections.

10.3 Explanatory scope

Integrated-information, global-workspace and predictive-processing accounts organize unity, access and modelling differently (Tononi et al., 2016; Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a candidate physical interaction and conditional relations between local content and self-encoding. Gluing and reconstruction permit multiple substrates; the electromagnetic commitment enters through physical calibration. That permission is wider than the unconditional results of Section 2 alone. The conditional chain is substrate-neutral in the same way: its field condition asks for a jointly continuous kernel on a normalized substrate, a coupling equal to that kernel’s double average, a noise that is the field’s own, and no single site carrying more than half the coupling—and none of these distinguishes an electromagnetic kernel from any other continuous mean-field kernel. The site condition is the only one with physical flavour, and it holds vacuously wherever the substrate has no atoms (`no_site_dominates`), which is the continuum case the electromagnetic hypothesis is about. The chain

therefore transfers to extracellular ion diffusion, to gap junctions modelled
 as a density, or to a mechanical or optical medium, modulo the calibration of
 Section 9.1. What carries the electromagnetic identification is that calibration
 together with the extracellular-geometry argument of Section 9, which is
 the one place a field equation enters. The formalization checks implications
 under their hypotheses, and its witnesses establish realizability on declared
 models. Two things stand outside that result and are not supplied by it: the
 eight hypotheses are discharged for toy components rather than for cortex,
 and the identification of the glued state with unity and of its reconstruction
 with a minimal self is a further commitment that the eight do not contain.

The explanatory question is whether the specified relations occur in neural
 activity and matter for experience. In the cup example, this means establishing
 how coordination helps maintain one represented situation and how the
 perceiver's place within it contributes to the episode. The proposal identifies
 candidate correlates and tests of that identification. Why those relations
 would be accompanied by experience remains unexplained.

Notation

Table 2: Symbols used throughout, with the section that defines each. Where one
 letter carries more than one meaning, the meanings are separated by the section
 they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i

Table 2 (continued)

Symbol	Defined	Meaning
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1)	Phase diffusion, in inverse time
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	Section 5	Number of communication rounds to a deadline
ν	Section 6	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (14)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (15)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph

Table 2 (continued)

Symbol	Defined	Meaning
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover’s nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
E, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s \mathcal{A})$
$\mathcal{E}$	Eq. (13)	Energy observable shared by the stages of a continuing protocol
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (15)	Field amplitude, in the cortical identification
U	§3.2, §4	Stored energy $U(\Xi)$ of a configuration; a patch U of the cover
s, s_*	§4–5	A global section and the one the reconstruction fixes

Data and code availability

The Lean development, simulation scripts, tests and saved numerical summaries are maintained in the GitHub repository `bobaseb/coupling_gated_cortical_coherence`. Publication simulation values are generated from saved summaries without rerunning production sweeps. The exploratory EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-processing and selection are described in Supplemental Material. The repository contains the reproducible analysis and does not redistribute the raw

1228 recordings.

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1230 This research received no external funding. The author declares no compet-
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1232 **Use of AI tools**

1233 AI assistance to this work was substantial and is not separable by task. Sev-
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1236 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
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1239 accessed with the Hermes CLI. Between them they contributed to the Lean
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1241 of both documents. Because the tools were used in combination and in alter-
1242 nation, no section, proof, figure or result is attributable to any one of them,
1243 and none is claimed to be. The author is responsible for all content, including
1244 everything a tool produced. Machine checking establishes the stated Lean
1245 results under their hypotheses; it does not validate biological interpretations
1246 or replace scientific review of generated code and prose.

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